Learning to Listen:

Scalable Species Detection from Environmental Audio Using Deep Convolutional Networks with Class-wise Attention Pooling and Curriculum Learning

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Abstract

Passive acoustic monitoring (PAM) produces far more audio than experts can annotate by hand, which motivates automatic species recognition from field recordings. We present a multi-label bioacoustic classifier for the BirdCLEF dataset, which covers birds, amphibians, insects, mammals and reptiles of the Brazilian Pantanal. Recordings are split into 5-second clips and turned into 3-channel log-mel/delta/delta-delta spectrograms. Each clip is encoded by a compact ResNet-style CNN, and a *class-wise attention pooling* layer combines the clips into one recording-level prediction. Training follows a three-phase curriculum that moves from a family-balanced, clean-first subset to the full dataset including soundscapes. The data is heavily imbalanced: 169 labelled classes, a median of 11 examples per class, a maximum of 666, and 67.7% of records are birds. On the internal validation split the model reaches a macro ROC-AUC of 0.5982, a macro PR-AUC of 0.0388, a rare-class ROC-AUC of 0.6236 and a family-level ROC-AUC of 0.6987. On the soundscape-style holdout test set these become 0.4770, 0.2544, 0.4661 and 0.4248. Below-chance holdout scores for head classes (0.3671), Amphibia (0.3333) and Mammalia (0.0243) point to a systematic domain shift between curated, bird-dominated focal recordings and amphibian-dominated soundscapes, not just weak learning. We report every result produced by the final run, analyse the model's attention behaviour, list the inconsistencies we found in the experimental record, and interpret what the results do and do not show.

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1 Introduction

Ecosystems worldwide face unprecedented pressure, and effective conservation increasingly depends on monitoring biodiversity at scale. In remote, ecologically rich regions, direct observation is costly and sparse. Passive acoustic monitoring offers an alternative: autonomous recorders listen to an ecosystem continuously and capture the vocal activity of many taxa over long periods. The bottleneck then moves from collecting data to *analysing* it. Thousands of hours of audio cannot be labelled manually, so automatic, scalable recognition methods are needed.

This study focuses on the Pantanal wetlands of Brazil, one of the largest and most biodiverse wetland systems on Earth. It covers more than 150,000 km^2 and is home to over 650 bird species and many other vocalising taxa. Much of the Pantanal remains under-monitored because of its scale, seasonal flooding, and growing threats from wildfire, agricultural expansion and climate change.

We study deep transfer-style learning for this setting using the BirdCLEF dataset, which combines focal recordings from citizen-science platforms (iNaturalist, Xeno-canto) with continuous soundscape recordings. These data are *weakly labelled*: a label applies to a whole recording, not to the moment a species calls, and several species may be present at once. The class distribution is also strongly long-tailed and taxonomically skewed towards birds.

Contributions.

1. A complete preprocessing and modelling pipeline that combines 3-channel spectral–temporal inputs, a lightweight ResNet-style clip encoder, and class-wise attention pooling for multiple-instance aggregation over clips (Section 4).
2. A three-phase curriculum (family-balanced → class-aware → natural with class-weighted loss) designed to counter class and taxonomic imbalance (Section 4.5).
3. A full report of all dataset statistics, overall, subgroup, family-level, training-dynamics and attention results (Sections 3–5).
4. An interpretation of the large validation-to-holdout gap, including below-chance AUCs, metric-prevalence effects and data-composition shift, plus an audit of inconsistencies in the experimental record (Sections 6–7).

2 Problem Statement

Let a recording r be split into T clips $X_r = \{x_1, \ldots, x_T\}$, and let $\mathbf{y}_r \in \{0,1\}^C$ be its multi-hot label vector over C classes. We learn $f : X_r \mapsto \mathbf{p}_r \in [0,1]^C$, where $p_{r,c}$ estimates the probability that class c is present *somewhere* in the recording. This is a multiple-instance learning (MIL) problem: the label is attached to the bag (recording), not the instances (clips).

Three properties make the problem hard:

- **Long-tailed classes.** Few classes have many examples; most have very few (Figure 1).
- **Taxonomic skew and domain shift.** Focal recordings are mostly birds, while soundscapes are mostly amphibians and insects (Section 3).
- **Weak and noisy labels.** Recording-level labels, optional and incomplete secondary labels, and quality ratings that vary widely.

Left unaddressed, imbalance makes a model learn the dominant bird classes well and the rare taxa poorly, which is the opposite of what biodiversity monitoring needs.

3 Data

3.1 Sources

We use the BirdCLEF data release, which has three parts: (i) `train.csv`, metadata for focal recordings in `train_audio/`; (ii) `train_soundscapes_labels.csv`, with semicolon-separated multi-species labels for continuous soundscapes; and (iii) `taxonomy.csv`, which maps each label to scientific name, common name and taxonomic class. `train.csv` has **35,549 rows and 15 columns**. Non-informative columns (`collection`, `author`, `license`, `url`, `inat_taxon_id`, `common_name`, `type`) were dropped. The remaining columns are primary label, secondary labels, latitude, longitude, scientific name, class name, rating, filename and a derived full path ($35{,}549 \times 9$; $35{,}549 \times 11$ after adding the parsed label list and encoded target). Secondary labels were parsed from their string form with `ast.literal_eval`. The full label set of a record is its primary label followed by its secondary labels, and the target is a multi-hot vector over the union of species in `train.csv` and the soundscape labels.

3.2 Quality groups

Each record is assigned to a *phase group* by its user rating: **clean** (rating $= 5$), **semi** ($2.5 \leq$ rating < 5), **noisy** (rating < 2.5), and **soundscape** (continuous field recordings). These groups drive the curriculum (Section 4.5).

3.3 Master metadata table

The curated master table (`master_df`) used for training contains **5,021 records** spanning **169 classes**. Table 1 summarises per-class label occurrences (primary and secondary combined). Figure 1 shows the histogram on a log scale.

Table 1: Per-class label-occurrence statistics in `master_df` (169 classes).

Statistic	Value
Number of classes	169
Mean examples per class	62.08
Standard deviation	120.06
Minimum	1
25th percentile	4
Median	11
75th percentile	59
Maximum	666
Implied total label occurrences (169×62.08)	$\approx$10,491
Mean / median ratio	5.6

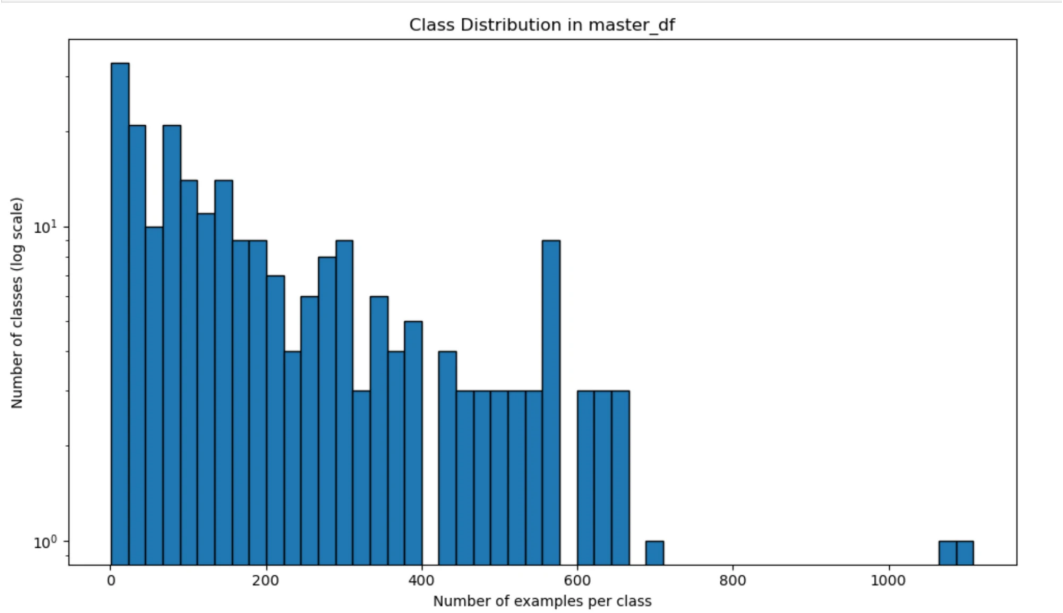


Figure 1: Log-scaled histogram of per-class example counts in `master_df`. Most classes sit in the first few bins (fewer than ∼100 examples), and a thin tail of head classes reaches several hundred examples. The distribution is strongly long-tailed.

3.4 Taxonomic composition

Table 2 gives record counts per taxonomic class ("family" in the code base), and Figure 2 shows the corresponding histogram.

Table 2: Records per taxonomic family in `master_df`.

Family	Records	Share
Aves	3,400	67.7%
Amphibia	1,249	24.9%
Insecta	247	4.9%
Mammalia	98	2.0%
Reptilia	27	0.5%
Total	5,021	100%

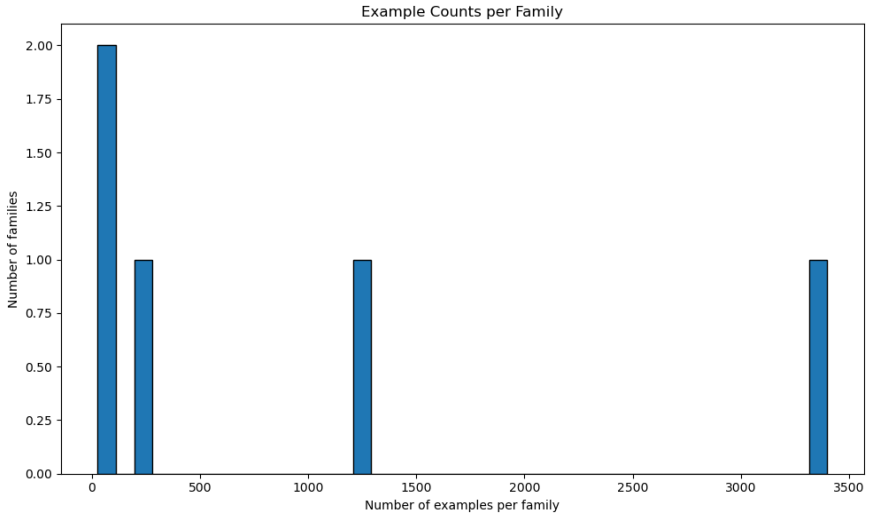


Figure 2: Histogram of example counts per family. Two families (Reptilia, Mammalia) have fewer than about 100 records, Insecta has about 250, Amphibia about 1,250, and Aves about 3,400. The largest family has 126× more records than the smallest.

3.5 Composition by quality group

Table 3 crosses quality group with family. This cross-tabulation explains much of the evaluation behaviour.

Table 3: Records per family within each quality group of `master_df`. Column totals sum to 5,021.

Family	Clean	Semi	Noisy	Soundscape
Aves	714	1,489	1,167	30
Amphibia	0	9	108	1,132
Insecta	0	0	7	240
Mammalia	3	1	44	50
Reptilia	0	0	1	26
Total	717	1,499	1,327	1,478
Share of all records	14.3%	29.9%	26.4%	29.4%

Two facts stand out. First, the *clean* subset is 99.6% birds (714 of 717) and contains no amphibian, insect or reptile recordings. Second, the *soundscape* subset is the reverse: 76.6% amphibian (1,132 of 1,478), 16.2% insect, and only 2.0% bird (30 records). Taxonomy and recording condition are therefore almost perfectly confounded.

3.6 Classes in the clean subset

Only **19 classes** appear in the clean group. Their label counts are listed in Table 4. All other classes are "missing from clean" and can only be learned from semi, noisy or soundscape audio.

Table 4: Label counts of the 19 classes present in the clean (rating = 5) subset of `master_df`.

#	Label	Count	#	Label	Count
1	saffin	82	11	whnjay1	35
2	rumfly1	77	12	grhtan1	33
3	gycwor1	71	13	purjay1	32
4	fepowl	65	14	rufcas2	19
5	rebscy1	63	15	blchaw1	14
6	greela	49	16	shshaw	6
7	redjun	47	17	ruther1	4
8	recfin1	41	18	47144	2
9	ragmac1	38	19	738183	1
10	yebela1	38			

3.7 Rating-5 recordings in the raw metadata

Over the full `train.csv`, the rating-5 recordings cover **173 primary-label classes**. Their per-class distribution is summarised in Table 5 and Figure 3. The smallest class has a single rating-5 recording, which confirms that even the highest-quality slice of the data is imbalanced.

Table 5: Samples per class among rating-5 recordings in `train.csv`.

Statistic	Value
Number of classes	173
Mean	39.57
Standard deviation	29.78
Minimum	1
25th percentile	18
Median	32
75th percentile	56
Maximum	142

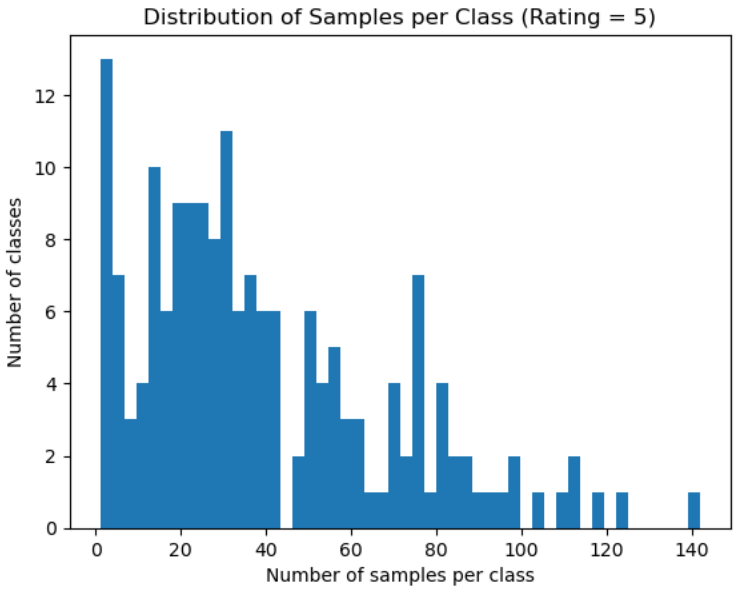


Figure 3: Distribution of samples per class among rating-5 recordings. The mode is at the very low end (13 classes in the first bin), and only a handful of classes exceed 100 samples.

4 Methods

4.1 Audio preprocessing

All recordings are converted to mono by averaging channels and resampled to 32,000 Hz. Each recording is split into non-overlapping 5-second clips,

$$X = \{x_1, x_2, \ldots, x_T\}, \qquad x_t \in \mathbb{R}^{160{,}000}, \tag{1}$$

with zero-padding on the final clip when needed. At most 8 clips per recording are used ($T \leq 8$).

4.2 Spectral representation

Each clip becomes a 3-channel stack of log-mel, delta and delta-delta spectrograms. The STFT is computed as

$$X[k,\tau] = \sum_{n} x[n]\, w[n - \tau H]\, e^{-j2\pi kn/N}, \qquad P[k,\tau] = |X[k,\tau]|^2, \tag{2}$$

with $N = n_{\text{fft}} = 2048$ and hop $H = 512$. It is projected onto 128 mel bands between $f_{\min} = 300$ Hz and $f_{\max} = 16{,}000$ Hz,

$$M[m,\tau] = \sum_{k} B[m,k]\, P[k,\tau], \qquad S[m,\tau] = 10 \log_{10}\big(\max(M[m,\tau], \epsilon)\big), \tag{3}$$

and resized to a fixed 128×384 resolution. The delta channels are first- and second-order temporal derivatives of S, computed with the usual regression formula

$$\Delta S[m,\tau] = \frac{\sum_{n=1}^{K} n\,(S[m,\tau+n] - S[m,\tau-n])}{2\sum_{n=1}^{K} n^2}, \qquad \Delta^2 S = \Delta(\Delta S). \tag{4}$$

Figure 4 shows an example. The log-mel channel shows harmonic banding (energy concentrated between bins ∼15–50 and a stationary band near bin 90–100). The delta channels emphasise onsets and transients as vertical striations, which makes temporal structure explicit to the CNN.

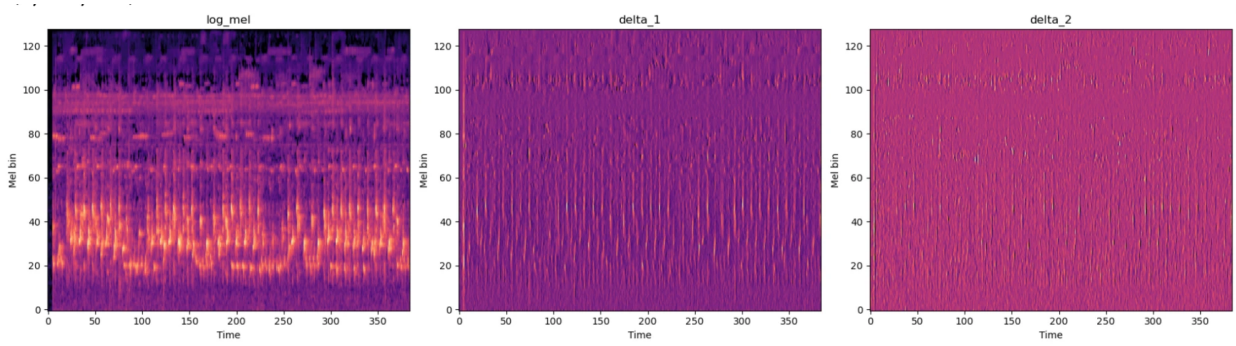


Figure 4: Example 3-channel model input (128 mel bins × 384 frames): log-mel spectrogram (left), first-order delta (centre) and second-order delta (right).

Normalisation. Each channel is standardised independently per frequency bin across time:

$$\mu_f = \frac{1}{T}\sum_{t=1}^{T} S_{f,t}, \quad \sigma_f = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(S_{f,t}-\mu_f)^2 + \epsilon}, \quad \tilde{S}_{f,t} = \frac{S_{f,t}-\mu_f}{\sigma_f}. \tag{5}$$

Per-bin normalisation removes stationary spectral colouring (microphone response, constant background hum), much like cepstral mean–variance normalisation in speech.

4.3 Clip encoder

Each clip $x_t \in \mathbb{R}^{3\times128\times384}$ is encoded by a ResNet-style CNN f_θ into an embedding $h_t \in \mathbb{R}^d$ with $d = 96$:

$$h_t = \mathrm{Dropout}_{0.2}(\mathrm{GAP}(F_\theta(x_t))). \tag{6}$$

The stem is a 7×7 stride-2 convolution, then batch normalisation, ReLU and 3×3 max pooling. Residual stages follow, each built from blocks of two 3×3 convolution–BN–ReLU layers with identity skip connections. Each stage starts with a strided convolution, so spatial resolution halves while channel width grows in the standard ResNet progression from a base width of 24. Global average pooling (GAP) over frequency and time produces the clip embedding. The 3-channel input lines up with RGB-shaped CNN stems and lets the first layer learn joint spectro-temporal filters.

4.4 Class-wise attention pooling

To turn clip embeddings into a recording-level prediction without timestamps, we use class-wise attention MIL pooling. For each class c, a small attention network g_c scores each clip:

$$e_{t,c} = g_c(h_t), \qquad \alpha_{t,c} = \frac{\exp(e_{t,c})}{\sum_{t'=1}^{T}\exp(e_{t',c})}, \qquad \sum_{t=1}^{T}\alpha_{t,c} = 1. \tag{7}$$

Padded clips are excluded through a clip mask. Each class gets its own pooled representation and logit:

$$z_c = \sum_{t=1}^{T}\alpha_{t,c}\,h_t, \qquad \hat{y}_c = \mathbf{w}_c^\top z_c + b_c, \qquad p_c = \sigma(\hat{y}_c) = \frac{1}{1+e^{-\hat{y}_c}}. \tag{8}$$

Unlike mean pooling, which weights every clip equally, or a single shared attention vector, the class-wise formulation lets different species depend on different parts of the same recording. In effect it acts as a weak event detector. The model also returns per-clip (instance) logits and probabilities, bag-level probabilities, clip and bag embeddings, and the attention matrix $\alpha \in \mathbb{R}^{T\times C}$, which we use for interpretation (Section 5.6).

$$\begin{array}{cccccc}
\text{Raw audio} & \rightarrow & \text{Mono, 32 kHz} & \rightarrow & \text{5-s clips } (\leq 8) & \rightarrow \\
\text{log-mel}/\Delta/\Delta^2 & \rightarrow & \text{Per-bin norm.} & \rightarrow & \text{ResNet encoder} & \rightarrow \\
h_t \in \mathbb{R}^{96} & \rightarrow & \text{class-wise attn. } \alpha_{t,c} & \rightarrow & z_c = \sum_t \alpha_{t,c} h_t & \rightarrow \\
\hat{y}_c = \mathbf{w}_c^\top z_c + b_c & \rightarrow & p_c = \sigma(\hat{y}_c) & & &
\end{array}$$

Figure 5: End-to-end model pipeline.

4.5 Training procedure: three-phase curriculum

Training follows a curriculum that moves from clean, balanced data to the full natural distribution.

Phase 1 (family-balanced). Target of 20 examples per family (five families, about 100 recordings). Examples are drawn clean-first: clean → semi → noisy → soundscape, the next group being used only when the previous one cannot meet the quota. Loss: unweighted BCE. Learning rate 10^{-3}.

Phase 2 (class-aware). Clean + semi + noisy data, sampled with class weights

$$w_c \propto \frac{1}{\sqrt{\text{class_count}_c}}, \tag{9}$$

a square-root inverse-frequency compromise between natural and fully balanced sampling. Loss: unweighted BCE. Learning rate 5×10^{-4}.

Phase 3 (natural + weighted loss). Full dataset including soundscapes, shuffled naturally. Loss: class-weighted BCE,

$$\mathcal{L} = \frac{1}{C} \sum_{c=1}^{C} \Big[-\beta_c \, y_c \log p_c - (1 - y_c) \log(1 - p_c) \Big], \tag{10}$$

where β_c up-weights positives of infrequent classes. Learning rate 10^{-4}.

Optimisation uses AdamW (weight decay 10^{-4}) with batch size 1 (one recording, i.e. one bag of up to 8 clips, per step). No data augmentation is used in the final run, so the effect of the curriculum and architecture can be seen in isolation.

4.6 Validation, stopping and model selection

A 10% internal validation split, stratified by soundscape presence where possible, is used for model selection and plateau detection. Each phase runs until validation macro ROC-AUC fails to improve by at least 0.001 for a fixed number of consecutive epochs (the experimental record states both 6 and 4; see Section 7). The checkpoint with the best validation macro ROC-AUC across all phases is kept. A separate holdout test set, closer to continuous soundscape recordings, is used only for final evaluation. All results come from a *single* final training run, not cross-validation.

4.7 Evaluation metrics

- **Macro ROC-AUC**: the mean over classes of the one-vs-rest ROC AUC, computed over classes with at least one positive example. Chance level is 0.5.
- **Macro PR-AUC**: the mean over classes of average precision. The chance level for class c equals its prevalence π_c, so the macro chance level depends on the evaluation set.
- **Rare-class ROC-AUC**: macro ROC-AUC over low-frequency classes.
- **Family-level ROC-AUC**: species probabilities are max-pooled within each family, $p_{\text{fam}} = \max_{c \in \text{fam}} p_c$, and ROC-AUC is computed per family.
- **Subgroups**: *head / medium / tail* (tertiles of training frequency), *rare*, *missing-from-clean* (absent from the clean subset), and *soundscape-heavy* ($\geq 50\%$ of samples from soundscapes).

4.8 Earlier pipeline iteration (repository code)

The repository also contains an earlier pipeline iteration, which is not the model behind the results reported here. That pipeline uses an ImageNet-pretrained ResNet-18 with a 1-channel stem and a BN–Dropout–Linear head (`BirdResNet`). Its log-mel settings are $n_{\text{fft}} = 1024$, hop 320, 128 mels over 20–14,000 Hz. It supports signal-centred, sequential, random, "WHEN"-detector and hybrid chunking, and it trains with BCE whose positive weights are $\text{neg}_c/\text{pos}_c$ clipped to $[1, 20]$. Its optional "WHEN" detector is a small CRNN (three conv–BN–ReLU blocks with 32/64/128 channels, frequency max-pooling, and a bidirectional GRU with hidden size 128) that predicts frame-level event activity. No evaluation results for this iteration are available, so it is described here only for completeness.

5 Results

5.1 Training setup

The final run used AdamW, batch size 1, encoder base width 24, embedding dimension 96, dropout 0.2, at most 8 clips per sequence, the three-phase curriculum, and no augmentation. The best checkpoint by validation macro ROC-AUC was kept.

5.2 Main results

Table 6 reports overall performance, and Figure 6 visualises it.

Table 6: Overall performance on the internal validation split and the holdout test set. Δ = holdout − validation.

Metric	Validation	Holdout Test	Δ
Macro ROC-AUC	0.5982	0.4770	−0.1212
Macro PR-AUC	0.0388	0.2544	+0.2156
Rare-class ROC-AUC	0.6236	0.4661	−0.1575
Family-level ROC-AUC	0.6987	0.4248	−0.2739

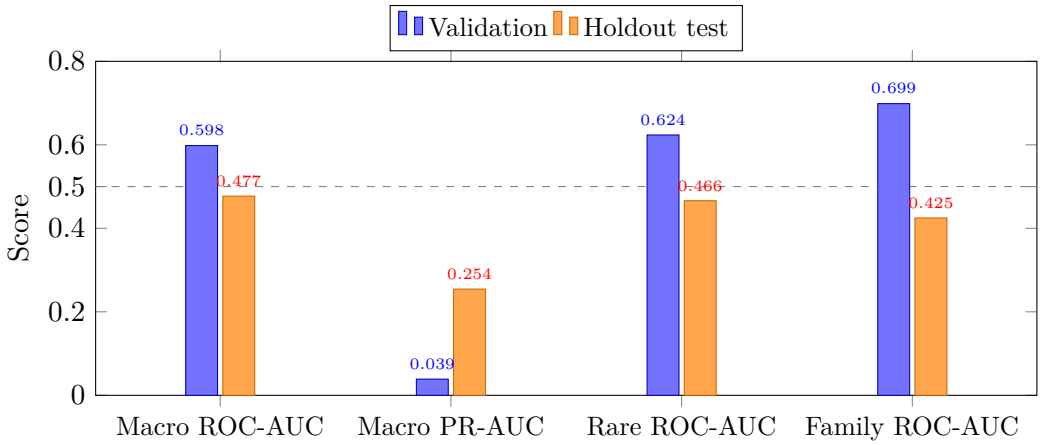


Figure 6: Overall metrics. The dashed line marks the ROC-AUC chance level (0.5); it does not apply to PR-AUC.

On validation, every ROC-based metric is above chance, and family-level discrimination (0.6987) is the strongest. On the holdout set, every ROC-based metric falls *below* 0.5, while macro PR-AUC rises by a factor of 6.6.

5.3 Subgroup analysis

Table 7 and Figure 7 break macro ROC-AUC down by class group.

Table 7: Macro ROC-AUC by class group.

Class group	Validation	Holdout Test	Δ
Head	0.5850	0.3671	−0.2179
Medium	0.6174	0.6935	+0.0761
Tail	0.6091	0.4661	−0.1430
Missing-from-clean	0.6407	0.4678	−0.1729
Soundscape-heavy	0.7333	0.4990	−0.2343

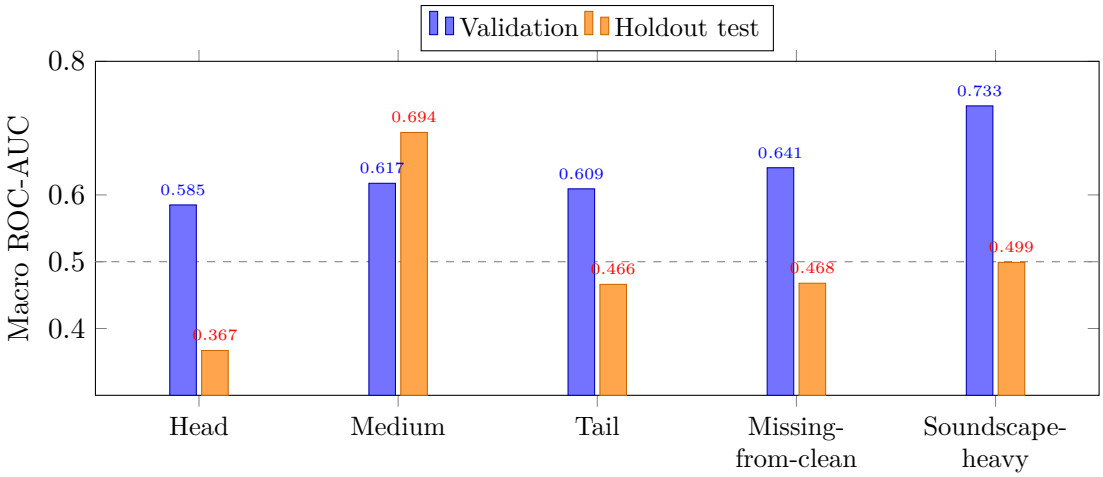


Figure 7: Subgroup macro ROC-AUC. Medium-frequency classes are the only group that improves on holdout data.

On validation the ordering is soundscape-heavy (0.7333) > missing-from-clean (0.6407) > medium (0.6174) > tail (0.6091) > head (0.5850): the least-frequent and least-curated groups score *best*. On holdout data the ordering becomes medium (0.6935) > soundscape-heavy (0.4990) > missing-from-clean (0.4678) > tail (0.4661) > head (0.3671). Medium-frequency classes are the only group that improves (+0.0761). Head classes fall well below chance.

5.4 Family-level results

Table 8 and Figure 8 report per-family ROC-AUC after max-pooling species probabilities within each family.

Table 8: Family-level ROC-AUC, ordered by validation score. Training record counts from Table 2 are repeated for reference.

Family	Validation	Holdout Test	Δ	Records
Reptilia	0.9494	0.7642	−0.1852	27
Insecta	0.8725	0.5081	−0.3644	247
Amphibia	0.8054	0.3333	−0.4721	1,249
Mammalia	0.5185	0.0243	−0.4942	98
Aves	0.3476	0.4942	+0.1466	3,400
Macro (all families)	0.6987	0.4248	−0.2739	

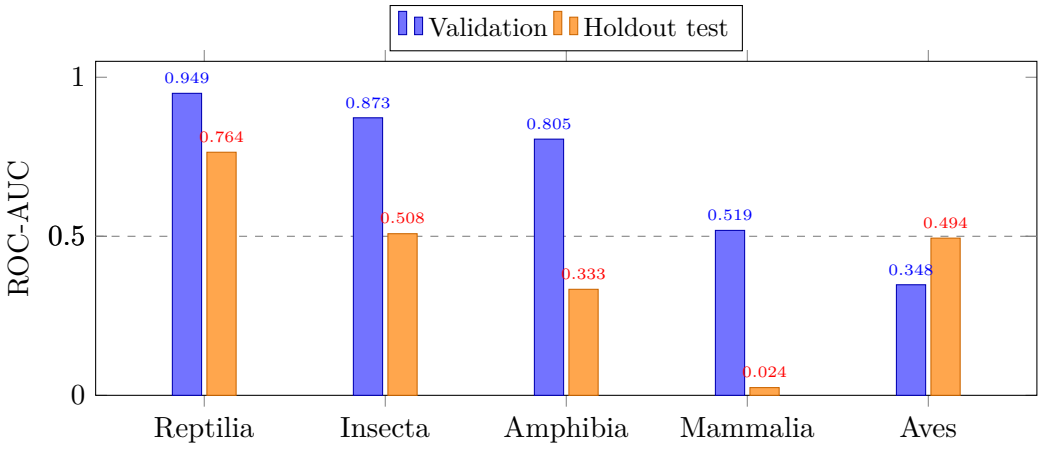


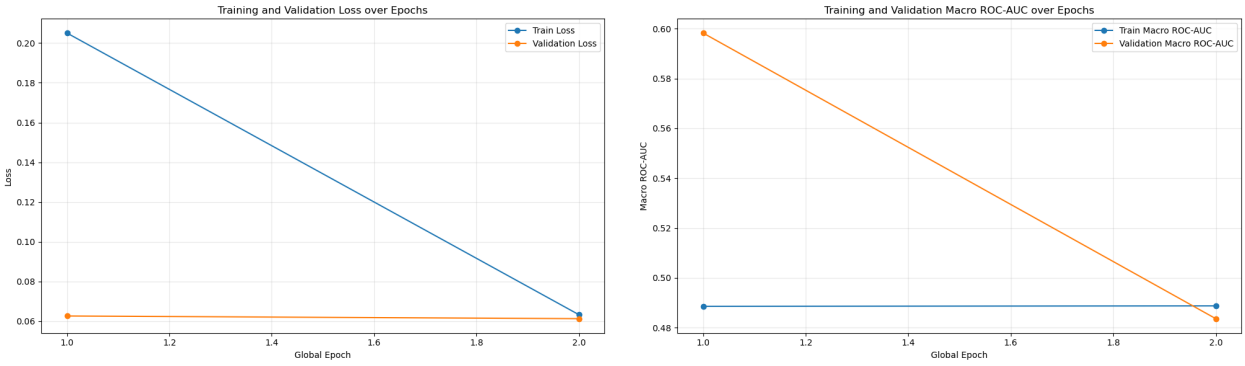
Figure 8: Family-level ROC-AUC. Aves is the only family that improves on holdout data. Mammalia collapses to almost perfectly inverted ranking (0.0243).

5.5 Training dynamics

The per-epoch history exported from the final run (`epoch_history.jsonl`) contains the two Phase-1 epochs in Table 9, plotted in Figures 9a and 9b.

Table 9: Per-epoch metrics recorded in `epoch_history.jsonl` (Phase 1, learning rate 10^{-3}).

Epoch	LR	Loss		Macro ROC-AUC		Macro PR-AUC	
		Train	Val	Train	Val	Train	Val
1	10^{-3}	0.205026	0.062613	0.488594	0.598242	0.080140	0.038834
2	10^{-3}	0.063286	0.061264	0.488779	0.483642	0.092219	0.039756
Δ		−0.141740	−0.001349	+0.000185	−0.114600	+0.012079	+0.000922



(a) Train and validation loss.

(b) Train and validation macro ROC-AUC.

Figure 9: Recorded training dynamics. Training loss falls by 69% between epochs 1 and 2 and converges onto the validation loss. Train ROC-AUC stays flat just below chance, and validation ROC-AUC drops from 0.598 to 0.484.

The run summary also records that the best-performing checkpoint was reached in Phase 3,

where validation macro ROC-AUC rose to **0.6783** before plateau-based stopping, which occurred after **24 epochs** in the final phase. Per-epoch values for Phases 2 and 3 are not included in the exported history. The relationship between the 0.6783 Phase-3 peak and the reported 0.5982 headline figure is discussed in Section 7.

5.6 Attention analysis

We ran the final checkpoint (`final_model.pt`) on recording `XC1001088`, whose ground-truth label is `rumfly1`. It yields two clips: clip_0 is a full 5-second window, and clip_1 contains only about 106 of 384 frames of real audio (about 1.4 s) followed by zero padding. The forward pass returns logits, probabilities, bag probabilities, clip embeddings, instance probabilities and logits, bag embeddings, attention weights and the clip mask. Figure 10 shows both spectrograms and the class-wise attention weights for the 8 highest-scoring classes. Table 10 lists the values.

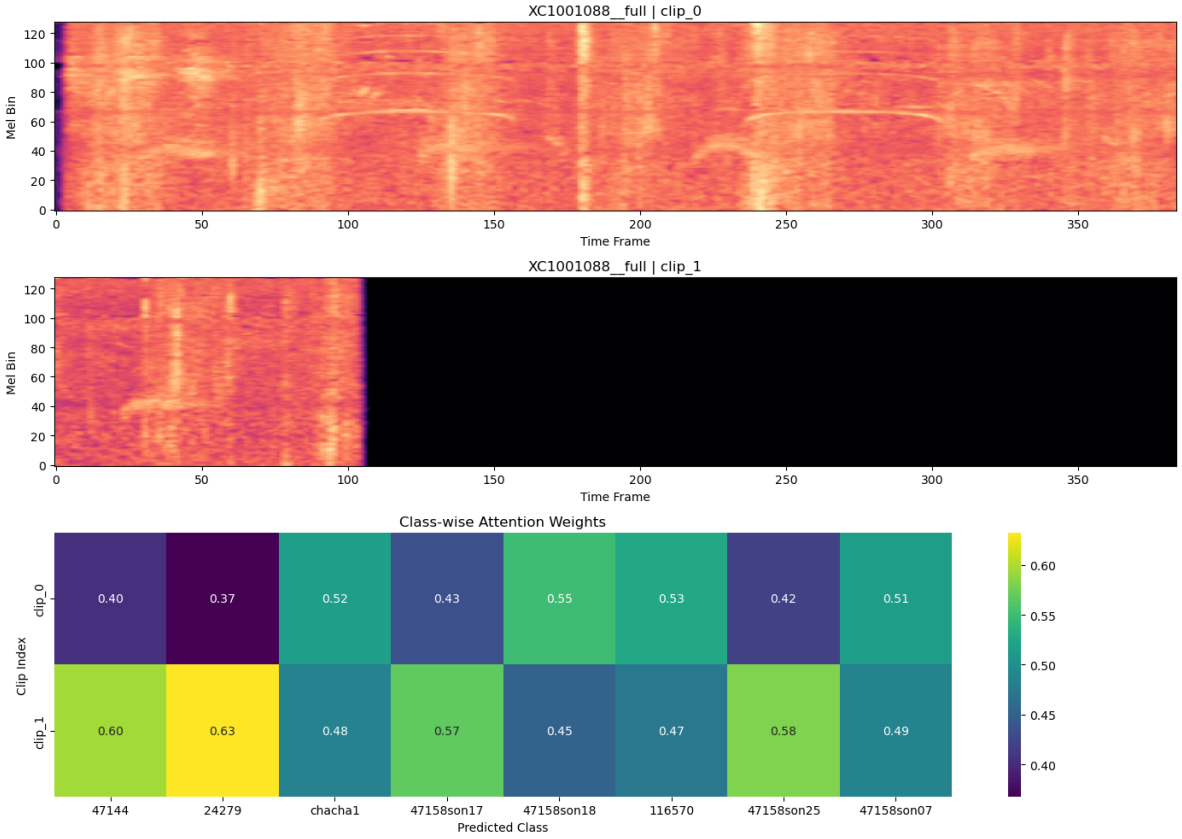


Figure 10: Class-wise attention for recording `XC1001088` (label `rumfly1`). Top: log-mel channel of clip_0 and clip_1. Bottom: attention weights $\alpha_{t,c}$ for the top-8 predicted classes.

Table 10: Attention weights $\alpha_{t,c}$ for the top-8 predicted classes on `XC1001088`, in descending order of bag probability. Each column sums to 1. "Preferred" marks the clip with the larger weight.

Class	47144	24279	chacha1	47158son17	47158son18	116570	47158son25	47158son07
clip_0	0.40	0.37	0.52	0.43	0.55	0.53	0.42	0.51
clip_1	0.60	0.63	0.48	0.57	0.45	0.47	0.58	0.49
Preferred	1	1	0	1	0	0	1	0
$\lvert\alpha_0 - \alpha_1\rvert$	0.20	0.26	0.04	0.14	0.10	0.06	0.16	0.02

Attention is split evenly across classes: four classes favour clip_0 and four favour clip_1, so the model does use the class-specific freedom of the pooling layer. All weights lie in a narrow 0.37–0.63 band, however, and the true label `rumfly1` is *not* among the top-8 predictions. These are mostly non-bird taxa: numeric iNaturalist IDs such as 116570 (the yacare caiman) and several `47158son*` insect sonotypes.

6 Interpretation

6.1 The model learns something, but much of it does not transfer

On validation, all ROC metrics sit clearly above 0.5, from 0.5982 macro to 0.6987 family-level, so the network extracts real discriminative information. On holdout data, every ROC metric falls below 0.5 (0.4770 macro, 0.4661 rare, 0.4248 family). A truly uninformative model would score about 0.5. Scores consistently *below* 0.5 mean the model ranks negatives above positives more often than chance would, so the cues it learned are *anti-correlated* with the labels on the holdout distribution. This is the signature of **shortcut learning under domain shift**, not of simple underfitting.

6.2 Taxonomy is confounded with recording condition

Table 3 suggests why. Birds are almost entirely focal recordings (clean, semi, noisy). Amphibians, insects and reptiles are almost entirely soundscapes. A model can reduce its training loss by learning "what a soundscape sounds like" (background noise floor, broadband stationary energy, recorder characteristics) as a proxy for "non-bird taxon", and "what a focal recording sounds like" (high-SNR foreground calls) as a proxy for "bird". Several results fit this reading:

- **Strong validation family scores for non-birds** (Reptilia 0.9494, Insecta 0.8725, Amphibia 0.8054) and for the soundscape-heavy group (0.7333). In the internal split, recognising the recording *condition* is enough to separate these families.
- **Weak validation score for Aves (0.3476).** When almost every non-soundscape recording is a bird, "is there a bird?" carries little information at the family level once probabilities are max-pooled over ∼hundreds of bird species. Max-pooling also inflates the family score of every recording, which hurts ranking.
- **On the soundscape-style holdout, the shortcut breaks.** When every holdout recording is a soundscape, the condition cue no longer separates families. Amphibia drops by 0.4721 and Insecta by 0.3644. Aves is the only family that *improves* (+0.1466 to 0.4942), which fits a model that had been actively penalising bird probabilities in soundscape-like audio. Removing that confound moves Aves back towards chance.

6.3 Below-chance extremes and small-sample variance

The holdout family values 0.3333 (Amphibia) and 0.0243 (Mammalia) deserve particular care. An ROC-AUC of exactly 1/3 is what one gets with very small positive/negative counts (for example one positive ranked against three negatives), and family-level AUC on a holdout of soundscape files is computed over relatively few recordings. Mammalia has only 98 training records in total and 50 in soundscapes. An AUC of 0.0243 means the ranking is almost perfectly *inverted*: flipping the scores would give 0.9757. The model does separate mammal-positive from mammal-negative holdout recordings, but in the wrong direction, which again points to a learned cue whose sign flips between domains. With a single run and no confidence intervals, these extremes should be read as strong qualitative evidence of shift, not as precise estimates.

6.4 Why PR-AUC goes *up* while ROC-AUC goes down

Macro PR-AUC rises from 0.0388 to 0.2544 even though ROC-AUC falls. This is not a contradiction. A random classifier's PR-AUC equals the positive prevalence. The validation split covers up to 169 classes with typically one label per recording (prevalence on the order of $1/169 \approx 0.006$ per class), so 0.0388 is several times chance. Soundscape holdout files are multi-label, and macro averaging covers only the subset of classes present, so prevalence per evaluated class is much higher and the chance level of PR-AUC rises with it. **PR-AUC values from the two sets therefore cannot be compared directly.** The holdout 0.2544 should be read against the holdout prevalence, and with ROC-AUC below 0.5 it is likely *at or below* the holdout chance level.

6.5 Imbalance handling: partial success on validation

On validation, the less-represented groups score better than head classes: rare 0.6236 and tail 0.6091 versus head 0.5850, and the rare-class score exceeds the overall macro score of 0.5982. This is the direction the curriculum was built to produce. Family-balanced Phase 1, square-root class-aware sampling in Phase 2 and positive-weighted BCE in Phase 3 all shift gradient mass towards tail classes. Within the training distribution, the imbalance countermeasures appear to work.

On the holdout set, however, tail (0.4661), rare (0.4661) and missing-from-clean (0.4678) classes all fall to just below chance, so the benefit does not survive the domain shift.

6.6 Why medium classes win and head classes lose on holdout

Medium-frequency classes are the only group to improve (0.6174 $\rightarrow$ 0.6935), while head classes collapse (0.5850 $\rightarrow$ 0.3671). The most frequent classes are mostly birds with large clean/semi focal collections (Table 4). These are the classes most exposed to the focal-recording shortcut, so their representations are tied to foreground, high-SNR conditions that do not occur in soundscapes. The positive weighting and class-aware sampling also push head-class logits *down* relative to the rest, which is useful under imbalance but harmful when those classes appear in a new domain. Medium-frequency classes have enough data to learn real acoustic patterns, yet are less dominated by any single recording condition, which gives them the best mix of signal and robustness. Tail classes simply have too few examples (median class size 11) to learn condition-invariant features.

6.7 What the training dynamics show

In the two recorded Phase-1 epochs, training loss falls sharply (0.205 $\rightarrow$ 0.063) and lands on the validation loss (0.061), while *train* ROC-AUC stays flat at about 0.489. A falling BCE with flat,

sub-chance ranking is typical of the early "predict-all-negative" phase of multi-label training: with roughly 169 mostly-zero targets per example, the model first learns to push every output towards the base rate, which minimises BCE without discriminating between classes. Validation ROC-AUC falling from 0.598 to 0.484 over the same step shows the epoch-1 value is not a stable estimate. Its size is plausible for an almost-untrained network evaluated on a small split, and it should be treated with caution. The reported Phase-3 peak of 0.6783 after 24 epochs shows that real discrimination emerges later, once the full dataset and weighted loss are used.

6.8 What the attention map shows

The attention example illustrates both the promise and the current weakness of class-wise pooling. The mechanism does what it is meant to do: different classes prefer different clips (four each way). Three warning signs appear:

1. **Low selectivity.** Weights range from 0.37 to 0.63, close to uniform (0.5). The model is not yet acting as a sharp event detector; in practice it is close to mean pooling.

2. **Attention to a mostly padded clip.** clip_1 is about 72% zero padding, yet it gets the *highest* weight in the map (0.63 for 24279) and is preferred by half the classes. A spectrogram region of padding, which after per-bin normalisation produces strongly atypical values, can act as a spurious feature. The final clip of nearly every recording is partly padded, so this is a likely systematic artefact.

3. **Wrong predictions.** The true species (`rumfly1`, a bird with 77 clean examples, the second-largest clean class) is not in the top 8. The top predictions are dominated by soundscape-heavy non-bird taxa. This matches the confounding hypothesis: the noisy-looking spectrogram of clip_0 (broadband vertical transients, diffuse background) resembles soundscape audio more than a clean focal recording.

6.9 Overall assessment

The final system is a well-motivated architecture whose training-distribution behaviour (above-chance ROC everywhere, strong family-level separation, tail classes outperforming head classes) shows the design choices have some effect. Its holdout behaviour shows that, as trained, it relies heavily on recording-condition cues that are confounded with taxonomy, and those cues reverse sign in deployment-like soundscapes. The main obstacle is not model capacity but **the confound between domain and label in the training data**, made worse by weak labels, no augmentation, and a single run with no uncertainty estimates.

7 Consistency Audit of the Experimental Record

To report results faithfully, we list the discrepancies found when comparing the run summary, the exported training history and the repository code.

1. **Headline metric versus best checkpoint.** The reported validation macro ROC-AUC (0.5982) and PR-AUC (0.0388) match the *epoch-1, Phase-1* values in `epoch_history.jsonl` (0.598242 and 0.038834) to the reported precision. The run summary also states that the best checkpoint came from Phase 3 at 0.6783. If selection maximised validation ROC-AUC across phases, the headline validation figure should be at least 0.6783. Either the headline validation

metrics were taken from the wrong row, or the evaluated checkpoint is the epoch-1 model. In the second case the holdout numbers describe an almost-untrained network. Which is true cannot be settled from the available artefacts, and it materially changes how the holdout numbers should be read.

2. **Incomplete history.** The exported history holds only two epochs, although Phase 3 alone is reported to have run for 24 epochs.

3. **Patience.** The methods description specifies a plateau patience of 6 epochs; the training-setup description specifies 4.

4. **Spectrogram settings.** The final run uses $n_{\text{fft}} = 2048$, hop 512, 300–16,000 Hz. The repository's `config.py` specifies $n_{\text{fft}} = 1024$, hop 320, 20–14,000 Hz, and the inference helper hard-codes hop 512 and $f_{\max} = 8000$ Hz. The class-wise attention model and its sequence dataset (`BirdCLEFSpectrogramModel`, `sequence_dataset`) are not in the repository, which contains only the earlier ResNet-18 pipeline.

5. **Class histogram range.** The summary statistics of per-class counts report a maximum of 666, whereas the class-distribution histogram (Figure 1) has non-empty bins near 1,100. The plotted call passes the whole two-column count table to the histogram, so the figure may not exactly match the statistics.

6. **Rare versus tail.** Holdout rare-class and tail ROC-AUC are identical (0.4661) while their validation values differ (0.6236 vs. 0.6091). This suggests the two groups coincide on the holdout set, or that one value was copied.

8 Limitations and Future Work

Limitations. (i) A single training run with no cross-validation or confidence intervals. (ii) No data augmentation. (iii) Weak recording-level labels with incomplete secondary labels. (iv) Severe confounding between taxonomy and recording condition. (v) A fixed 5-second clip representation with at most 8 clips, so long-range temporal context is lost and the last clip is often mostly padding. (vi) The inconsistencies listed in Section 7.

Future work.

- **Fix and re-evaluate.** Re-export the full per-epoch history, confirm which checkpoint was evaluated, re-run holdout evaluation with the Phase-3 best checkpoint, and report bootstrap confidence intervals and per-class positive counts.
- **Break the confound.** Mix soundscape background into focal bird recordings and focal non-bird calls into clean backgrounds, add background-noise injection, SpecAugment-style time/frequency masking and mixup, so that recording condition stops predicting taxonomy.
- **Mask padding.** Exclude padded frames from normalisation and attention, or replace zero padding with loop or reflection padding.
- **Sharper attention.** Use attention temperature or entropy regularisation, or auxiliary instance-level losses, to encourage selective event localisation.

- **Domain adaptation and pseudo-labelling.** Use unlabelled soundscapes through self-training or adversarial domain alignment; evaluate with prevalence-aware metrics such as padded cmAP.
- **Stronger encoders.** Use pretrained audio backbones and sequence models over clip embeddings to capture longer context.

9 Conclusion

We built a CNN-based bioacoustic classifier with 3-channel spectral–temporal inputs, class-wise attention pooling and a three-phase imbalance-aware curriculum for the long-tailed, multi-taxon BirdCLEF Pantanal dataset. On internal validation it reaches a macro ROC-AUC of 0.5982 (0.6783 Phase-3 peak), a rare-class ROC-AUC of 0.6236, a family-level ROC-AUC of 0.6987, and per-family scores up to 0.9494 (Reptilia), with tail classes outperforming head classes. On the soundscape-style holdout set, performance falls to a macro ROC-AUC of 0.4770, rare-class 0.4661 and family-level 0.4248. Only medium-frequency classes (0.6935) and Aves (+0.1466) improve. The systematic below-chance scores, together with the near-total confounding of taxonomy and recording condition in the training data, point to domain shift and shortcut learning as the main failure mode. Closing the gap between curated recordings and real soundscapes, through condition-breaking augmentation, padding-aware attention, domain adaptation and a careful re-evaluation of the final checkpoint, is the clearest path to a deployable monitoring system.

Reproducibility Summary

Item	Setting
Audio	mono, 32 kHz, non-overlapping 5 s clips, zero-padded, $\leq$8 clips/recording
Features	log-mel + Δ + Δ^2; $n_{\text{fft}} = 2048$, hop 512, 128 mels, 300–16,000 Hz
Input size	$3 \times 128 \times 384$, per-channel per-frequency-bin standardisation
Encoder	ResNet-style, base width 24, GAP, embedding $d = 96$, dropout 0.2
Pooling	class-wise attention MIL with clip mask
Curriculum	P1: 20/family, clean-first, BCE, LR 10^{-3}; P2: $w_c \propto c_c^{-1/2}$, BCE, LR 5×10^{-4}; P3: all data incl. soundscapes, class-weighted BCE, LR 10^{-4}
Optimiser	AdamW, weight decay 10^{-4}, batch size 1, no augmentation
Selection	10% validation split; plateau $\Delta < 0.001$ (patience 4 or 6); best val macro ROC-AUC
Evaluation	macro ROC-AUC, macro PR-AUC, rare, family (max-pooled), head/medium/tail, missing-from-clean, soundscape-heavy; single run

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